

Engineering Mathematics – I

Complex Numbers · Probability & Statistics · Integral Calculus · Numerical Methods & Vector Algebra

Complete handbook for the Kerala Diploma Revision 2026 curriculum. Covers all four modules with deep explanations, worked examples, interactive calculators, Malayalam support, and exam-oriented practice.

PROGRAMME	Common to all Engineering Diploma programmes
COURSE TYPE	Theory
CONTACT HOURS	60 (L:45 + T:15)
SELF-LEARNING	60 hours (suggested)
CIE / ESE	40 / 60

MODULE I	Complex Numbers (14 h)
MODULE II	Probability & Statistics (15 h)
MODULE III	Integral Calculus (15 h)
MODULE IV	Numerical Methods & Vector Algebra (16 h)



Official Source Declaration

Source: Kerala Polytechnic Revision 2026 Syllabus — Course 2001A (Engineering Mathematics – I).

This handbook is an educational study aid developed for students. It is not an official publication of the examining body.

⚠ Verified Inconsistencies: The syllabus lists Module IV self-learning activities with a typo — "simpson's $\left(\frac{1}{2}\right)$ rd rule" — which is corrected to **Simpson's 1/3rd rule** throughout this handbook, matching the main syllabus entry. The total self-learning hours sum to 60 as stated. All other data reconcile.



Course Dashboard

60

CONTACT HOURS

4

CREDITS

40+60

CIE / ESE

4

MODULES



Prerequisites

Fundamentals of Engineering Mathematics I (Course 1002) — Trigonometry, Determinants, Differentiation.

Teaching & Assessment Scheme

Teaching Scheme	Self-Learning/Week	Total Hours	Credits	CIE	ESE
3:1:0:0 (L:T:P:R)	4 h	60	4	40	60

Module-wise Instructional Hours				
Module	Title	Lecture (L)	Tutorial (T)	Total
I	Complex Numbers	10	4	14
II	Probability & Statistics	12	3	15
III	Integral Calculus	11	4	15
IV	Numerical Methods & Vector Algebra	12	4	16
Total			60	

How to Use This Handbook

1

Understand

Read each module's explanation, formulae, and worked examples.

2

Visualise

Study the SVG diagrams and interactive calculators.

3

Apply

Solve the module-wise questions and try the model paper.

4

Revise

Use the Quick Revision cards and glossary before exams.

Course Outcomes

CO1

Apply concepts of complex numbers to solve problems. (Bloom: Apply)

CO2

Apply concepts of Probability & Statistics in various mathematical problems. (Bloom: Apply)

CO3

Use integration techniques and differential equations to solve basic engineering problems. (Bloom: Apply)

CO4

Apply basic numerical methods and vector algebra techniques to solve simple engineering problems. (Bloom: Apply)

CO–PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-

Legends: 1–Low; 2–Medium; 3–High; – No Correlation



Curriculum Coverage Map

Module → Topics → Hours → CO → Handbook Anchor

Module	Major Topics	Hours	CO	Section
I	Complex number system, operations, Argand plane, modulus, amplitude, polar form, problems	14	CO1	Module 1
II	Sets, probability axioms, coins/dice/cards, statistics, frequency distribution, mean/median/mode	15	CO2	Module 2
III	Integration as reverse, standard integrals, substitution, parts, definite integrals, differential equations	15	CO3	Module 3
IV	Interpolation, Lagrange, Simpson's 1/3rd, scalars/vectors, dot/cross product, magnitude, unit vector	16	CO4	Module 4

Coverage note: The differential-equations topic includes the first-order *linear differential equation* form $dy/dx + P_y = Q$, together with variable-separable equations.

How the Modules Connect

Module 1

Complex Numbers

Foundation for AC circuit analysis, signal processing, and control systems.

Module 2

Probability & Stats

Data analysis, quality control, risk assessment, and decision-making.

Module 3

Integral Calculus

Area, volume, work, and modelling dynamic systems via differential equations.

Module 4

Numerical Methods & Vectors

Approximation, numerical integration, and spatial/physical vector analysis.

MODULE I **Complex Numbers**

14 hours (L:10 + T:4)

CO1

Bloom: Apply

Learning Goals

- Understand the complex number system
- Perform fundamental operations
- Represent complex numbers graphically
- Find modulus, amplitude, and polar form
- Solve engineering problems

Key Facts

- $i = \sqrt{-1}$
- $z = a + ib$
- $|z| = \sqrt{a^2 + b^2}$
- $\arg(z) = \tan^{-1}(b/a)$
- Polar: $z = r(\cos \theta + i \sin \theta)$

The Complex Number System

A **complex number** is a number of the form $z = a + ib$, where **a** and **b** are real numbers, and **i** is the imaginary unit defined by $i^2 = -1$.

- **Real part:** $\text{Re}(z) = a$
- **Imaginary part:** $\text{Im}(z) = b$

$$z = a + ib \cdot a, b \in \mathbb{R} \cdot i^2 = -1$$

Example: $z = 3 + 4i$ has $\text{Re} = 3$ and $\text{Im} = 4$.

Malayalam support: കോംപ്ലക്സ് നമ്പർ (complex numbers) $a + ib$ രൂപയിൽ എഴുതാം. ഇവിടെ 'i' $\sqrt{-1}$ നെ സൂചിപ്പിക്കുന്നു. $i^2 = -1$.

 **Tip:** The real and imaginary parts are always real numbers. The imaginary part includes the coefficient of i, not i itself.

Conjugate and Equality

Conjugate: The conjugate of $z = a + ib$ is $\bar{z} = a - ib$.

$$\bar{z} = a - ib$$

Equality: Two complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are equal iff $a_1 = a_2$ **and** $b_1 = b_2$.

Example: If $2x + 3iy = 4 + 6i$, then $2x = 4 \Rightarrow x = 2$, and $3y = 6 \Rightarrow y = 2$.



Tip: The conjugate is used to rationalise denominators and find modulus.

Fundamental Operations

For $z_1 = a + ib$ and $z_2 = c + id$:

- **Addition:** $z_1 + z_2 = (a+c) + i(b+d)$
- **Subtraction:** $z_1 - z_2 = (a-c) + i(b-d)$
- **Multiplication:** $z_1 \cdot z_2 = (ac - bd) + i(ad + bc)$
- **Division:** $z_1 / z_2 = (a+ib)/(c+id) \times (c-id)/(c-id) = [(ac+bd) + i(bc-ad)] / (c^2 + d^2)$

Example (Multiplication): $(2+3i)(1-4i) = 2 - 8i + 3i - 12i^2 = 2 - 5i + 12 = 14 - 5i$

Example (Division): $(3+4i)/(1-2i) = (3+4i)(1+2i)/5 = (3+6i+4i+8i^2)/5 = (-5+10i)/5 = -1 + 2i$



Common mistake: In division, always multiply numerator and denominator by the conjugate of the denominator.

Graphical Representation — Argand Plane

Complex numbers are represented on the **Argand plane** (complex plane) with the real part on the horizontal axis and the imaginary part on the vertical axis.

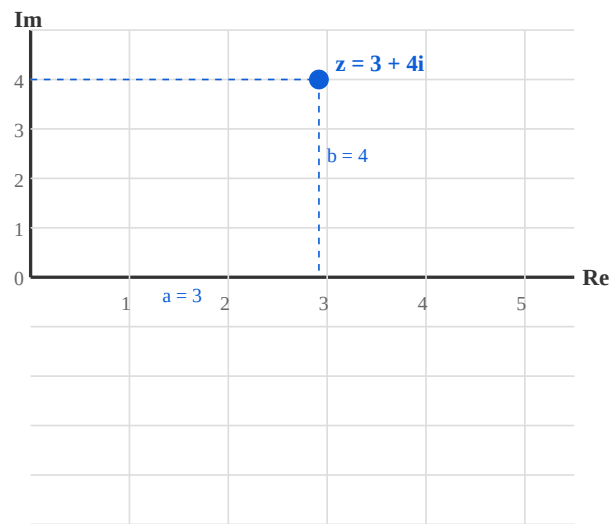


Figure 1: Argand plane showing $z = 3 + 4i$ with real part $a = 3$ and imaginary part $b = 4$.

Malayalam support: Argand plane- complex numbers- points represent . Horizontal axis real part-um vertical axis imaginary part-um .

Modulus and Amplitude

Modulus (Absolute value): $|z| = \sqrt{a^2 + b^2}$. It is the distance from the origin to the point (a, b) .

Amplitude (Argument): $\theta = \arg(z) = \tan^{-1}(b/a)$, measured in degrees (or radians).

$$|z| = \sqrt{a^2 + b^2} \quad \cdot \quad \theta = \tan^{-1}(b/a)$$

Example: For $z = 3 + 4i$: $|z| = \sqrt{9+16} = 5$, $\theta = \tan^{-1}(4/3) \approx 53.13^\circ$

! Common mistake: The argument depends on the quadrant. Always check the signs of a and b . In QII, $\theta = 180^\circ - \tan^{-1}(b/|a|)$.

Polar Form

A complex number can be expressed in **polar form** as:

$$z = r(\cos \theta + i \sin \theta) \cdot r = |z|, \theta = \arg(z)$$

This is also written as $z = r \angle \theta$ or $z = r \operatorname{cis} \theta$.

Example: For $z = 3 + 4i$: $r = 5$, $\theta = 53.13^\circ \Rightarrow z = 5(\cos 53.13^\circ + i \sin 53.13^\circ)$

 **Tip:** Polar form is especially useful for multiplication and division — multiply moduli and add arguments.

Solved Problems

Problem 1: Find the modulus and argument of $z = -1 + i$.

Solution: $a = -1$, $b = 1$. $r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$. $\theta = 180^\circ - \tan^{-1}(1/1) = 180^\circ - 45^\circ = 135^\circ$.

Problem 2: Express $z = 2 - 2i$ in polar form.

Solution: $r = \sqrt{2^2 + (-2)^2} = 2\sqrt{2}$. $\theta = -\tan^{-1}(2/2) = -45^\circ$. $z = 2\sqrt{2}(\cos(-45^\circ) + i \sin(-45^\circ))$.

Problem 3: If $z_1 = 1+2i$ and $z_2 = 3-i$, find $z_1 \times z_2$.

Solution: $(1+2i)(3-i) = 3 - i + 6i - 2i^2 = 3 + 5i + 2 = 5 + 5i$.

 **Engineering Application:** Complex numbers model AC circuits — impedance $Z = R + iX$, with magnitude giving total opposition and phase angle giving phase shift.



Complex Number Calculator

Enter real and imaginary parts to compute modulus, argument, and polar form.

Real part (a)

Imaginary part (b)

3

4

$z = 3 + 4i$ | $|z| = 5.000$ | $\arg = 53.13^\circ$ | Polar: $5.000(\cos 53.13^\circ + i \sin 53.13^\circ)$

MODULE II Probability & Statistics

15 hours (L:12 + T:3)

CO2

Bloom: Apply



Learning Goals

- Understand sets, random experiments, sample space
- Apply classical probability axioms
- Solve problems with coins, dice, cards
- Organise data and compute mean/median/mode



Key Facts

- $P(E) = n(E)/n(S)$
- $0 \leq P(E) \leq 1$
- Mean = $\Sigma x/n$
- Median = middle value (ordered)
- Mode = most frequent value

Sets: Empty, Universal, Subsets

- **Set:** A well-defined collection of objects.
- **Empty set ($\emptyset$):** A set with no elements.
- **Universal set (U):** The set containing all elements under consideration.
- **Subset:** $A \subseteq B$ if every element of A is also in B.

Example: For rolling a die, $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 4, 6\}$ (even numbers) is a subset of U.

Malayalam support: Set എന്നത് നിശ്ചിതമായ വസ്തുക്കളുടെ ശേഖരമാണ്. Empty set-ന് എന്തും ഉണ്ടാകില്ല. Universal set-ന് എന്തും ഉണ്ടാകാം.

Random Experiments, Outcomes, Sample Space

- **Random experiment:** An experiment whose outcome is not certain.
- **Outcome:** A single result of an experiment.
- **Sample space (S):** The set of all possible outcomes.
- **Event (E):** A subset of the sample space.

Example: Tossing a coin: $S = \{H, T\}$. Event $E = \{H\}$ (getting heads).

Classical Definition & Axioms of Probability

Classical definition: For a random experiment with $n(S)$ equally likely outcomes, the probability of event E is:

$$P(E) = n(E) / n(S)$$

Axioms:

- $0 \leq P(E) \leq 1$
- $P(S) = 1$
- For mutually exclusive events, $P(E_1 \cup E_2) = P(E_1) + P(E_2)$



Tip: The sum of probabilities of all outcomes in S is 1.

Problems: Coins, Die, Cards

Coin (1 coin): $S = \{H, T\}$. $P(H) = 1/2$.

Coin (2 coins): $S = \{HH, HT, TH, TT\}$. $P(HT) = 1/4$, $P(\text{at least one H}) = 3/4$.

Die: $S = \{1, 2, 3, 4, 5, 6\}$. $P(\text{even}) = 3/6 = 1/2$.

Cards: From a deck of 52, $P(\text{heart}) = 13/52 = 1/4$.

 **Common mistake:** In "at least one" problems, use the complement rule: $P(\text{at least one}) = 1 - P(\text{none})$.

Introduction to Statistics

Statistics: The science of collecting, organising, analysing, and interpreting data.

- **Ungrouped data:** Raw data as collected.
- **Frequency distribution:** A table showing how often each value occurs.

Example: Marks: 5, 7, 7, 8, 9. Frequency: 5→1, 7→2, 8→1, 9→1.

Measures of Central Tendency

Mean ($\bar{x}$): Sum of all values $\div$ number of values.

$$\bar{x} = \Sigma x / n$$

Median: The middle value when data is ordered (or average of two middle values if n is even).

Mode: The value that occurs most frequently.

Example: Data: 3, 5, 7, 7, 9. Mean = $31/5 = 6.2$. Median = 7. Mode = 7.

Example (even n): 2, 4, 6, 8. Median = $(4+6)/2 = 5$.



Tip: Mean is affected by outliers; median and mode are more robust.

Frequency Distribution — Mean, Median, Mode

For grouped frequency distribution:

$$\bar{x} = \Sigma(f \cdot x) / \Sigma f$$

Median for grouped data: find the class containing $(n+1)/2$ -th observation.

Mode for grouped data: the class with highest frequency.

Example: x : 1, 2, 3, 4 f : 2, 3, 1, 2. $\Sigma f = 8$. $\Sigma(f \cdot x) = 2+6+3+8 = 19$. $\bar{x} = 19/8 = 2.375$.

Malayalam support: Frequency distribution- $\bar{x}$ data- $\bar{x}$ groups $\bar{x}$ $\bar{x}$ frequencies $\bar{x}$ $\bar{x}$. Mean, median, mode $\bar{x}$ central tendency- $\bar{x}$ measures $\bar{x}$.



Mean, Median, Mode Calculator

Enter numbers separated by commas (e.g. 5,7,7,8,9).

Data

5,7,7,8,9

Mean: 7.200 | Median: 7.000 | Mode: 7

MODULE III Integral Calculus

15 hours (L:11 + T:4)

CO3

Bloom: Apply



Learning Goals

- Understand integration as reverse of differentiation
- Recall standard integrals
- Apply substitution and integration by parts
- Evaluate definite integrals
- Solve first-order differential equations



Key Facts

- $\int x^n dx = x^{n+1}/(n+1) + C$ ($n \neq -1$)
- $\int e^x dx = e^x + C$
- $\int 1/x dx = \ln|x| + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$

Integration as Reverse of Differentiation

Integration is the inverse process of differentiation. If $\frac{d}{dx} [F(x)] = f(x)$, then $\int f(x) dx = F(x) + C$, where C is the constant of integration.

$$\int f(x) dx = F(x) + C \quad \cdot \quad \frac{d}{dx} [F(x)] = f(x)$$

Example: $\frac{d}{dx} (x^2) = 2x \Rightarrow \int 2x dx = x^2 + C.$

 **Tip:** The constant C accounts for all antiderivatives. Always include it in indefinite integrals.

Standard Integrals

$$\int x^n dx = x^{n+1}/(n+1) + C$$

$n \neq -1$

$$\int 1/x dx = \ln|x| + C$$

$x \neq 0$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

Rules of Integration & Substitution

Rules:

- $\int k \cdot f(x) \, dx = k \cdot \int f(x) \, dx$
- $\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$

Substitution:

- $\int f(ax+b) \, dx = (1/a) F(ax+b) + C$
- $\int f(x) \cdot f'(x) \, dx = [f(x)]^2/2 + C$
- $\int f'(x)/f(x) \, dx = \ln|f(x)| + C$

Example (substitution): $\int (2x+3)^5 \, dx$. Let $u = 2x+3$, $du = 2dx \Rightarrow dx = du/2$.
 $= 1/2 \int u^5 \, du = u^6/12 + C = (2x+3)^6/12 + C$.

Example (f'/f): $\int (2x)/(x^2+1) \, dx = \ln|x^2+1| + C$.

Integration by Parts

For product of two functions:

$$\int u \, dv = uv - \int v \, du$$

LIATE rule (choose u as the function that comes first): Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential.

Example: $\int x \sin x \, dx$. Let $u = x$ (algebraic), $dv = \sin x \, dx \Rightarrow du = dx$, $v = -\cos x$.
 $= x(-\cos x) - \int (-\cos x) \, dx = -x \cos x + \sin x + C$.

⚠ Common mistake: Choosing the wrong u can make the integral more complex. Use LIATE.

Definite Integrals

A **definite integral** has limits of integration:

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

Example: $\int_0^1 x^2 dx = [x^3/3]_0^1 = 1/3 - 0 = 1/3.$

Example (area): Area bounded by $y = x^2$, x-axis, $x=0$, $x=2$ is $\int_0^2 x^2 dx = [x^3/3]_0^2 = 8/3$ sq. units.



Engineering Application: Definite integrals compute areas, volumes, work done, and centre of mass.

Differential Equations

Order: The highest derivative present.

Degree: The exponent of the highest derivative (after removing radicals).

Example: $d^2y/dx^2 + 3(dy/dx) + 2y = 0 \rightarrow$ Order 2, Degree 1.

Variable separable: $dy/dx = f(x)g(y) \Rightarrow \int dy/g(y) = \int f(x) dx.$

Example: $dy/dx = xy \Rightarrow dy/y = x dx \Rightarrow \ln|y| = x^2/2 + C \Rightarrow y = C e^{(x^2/2)}.$

Linear first-order: $dy/dx + P(x)y = Q(x).$

Integrating factor: $I.F. = e^{(\int P dx)} \cdot y \cdot I.F. = \int Q \cdot I.F. dx + C$

Example: $dy/dx + 2y = x.$ $P = 2$, $I.F. = e^{(\int 2 dx)} = e^{(2x)}.$ $y e^{(2x)} = \int x e^{(2x)} dx + C.$ Solve to get $y = (2x-1)/4 + C e^{(-2x)}.$

Definite Integral Calculator (polynomial)

Compute $\int_a^b x^n dx$. Enter a, b, and n.

a (lower limit)

0

b (upper limit)

2

n (exponent)

2

$$\int_a^b x^n dx = 2.666667 \text{ (a=0, b=2, n=2)}$$

MODULE IV Numerical Methods & Vector Algebra

16 hours (L:12 + T:4)

CO4

Bloom: Apply

Learning Goals

- Understand interpolation and Lagrange's method
- Apply Simpson's 1/3rd rule
- Distinguish scalars and vectors
- Compute dot and cross products
- Find magnitude and unit vectors

Key Facts

- Lagrange interpolation: $L(x) = \sum y_i \cdot \ell_i(x)$
- Simpson's 1/3rd: $\int_a^b f(x)dx \approx h/3 [f_0 + f_n + 4\sum_{\text{odd}} + 2\sum_{\text{even}}]$
- $a \cdot b = |a||b| \cos \theta$
- $a \times b = |a||b| \sin \theta \cdot \hat{n}$

Interpolation & Lagrange's Method

Interpolation is the process of estimating unknown values between known data points.

Lagrange's method constructs a polynomial that passes through all given points:

$$L(x) = \sum_{i=0}^n y_i \cdot \ell_i(x), \text{ where } \ell_i(x) = \prod_{j=0, j \neq i}^n (x - x_j) / (x_i - x_j)$$

Example: Given (1,2), (2,3), (3,5). Find y at x=2.5.

$$\ell_0 = (x-2)(x-3)/((1-2)(1-3)) = (x-2)(x-3)/2$$

$$\ell_1 = (x-1)(x-3)/((2-1)(2-3)) = -(x-1)(x-3)$$

$$\ell_2 = (x-1)(x-2)/((3-1)(3-2)) = (x-1)(x-2)/2$$

$$L(2.5) = 2 \cdot \ell_0 + 3 \cdot \ell_1 + 5 \cdot \ell_2. \text{ Compute: } \ell_0 = (0.5)(-0.5)/2 = -0.125, \\ \ell_1 = -(1.5)(-0.5) = 0.75, \ell_2 = (1.5)(0.5)/2 = 0.375. L = \\ 2(-0.125) + 3(0.75) + 5(0.375) = -0.25 + 2.25 + 1.875 = 3.875.$$



Tip: Lagrange's method avoids solving systems of equations but can be computationally intensive for large n.

Numerical Integration — Simpson's 1/3rd Rule

For approximating $\int_a^b f(x) dx$ with n even subintervals of width $h = (b-a)/n$:

$$\int_a^b f(x) dx \approx (h/3) [f_0 + f_n + 4(f_1 + f_3 + \dots + f_{n-1}) + 2(f_2 + f_4 + \dots + f_{n-2})]$$

Example: Approximate $\int_0^2 x^2 dx$ using n=4. $h = 0.5$. $f_0=0$, $f_1=0.25$, $f_2=1$, $f_3=2.25$, $f_4=4$. $\approx (0.5/3)[0+4 + 4(0.25+2.25) + 2(1)] = (1/6)[4 + 4(2.5) + 2] = (1/6)[4+10+2] = 16/6 = 2.6667$. Exact: $8/3 = 2.6667$.



Common mistake: n must be even for Simpson's 1/3rd rule. If n is odd, use Simpson's 3/8th rule or composite methods.

Scalars and Vectors

- **Scalar:** A quantity with only magnitude (e.g., mass, temperature).
- **Vector:** A quantity with both magnitude and direction (e.g., velocity, force).

Unit vectors: $\hat{i}, \hat{j}, \hat{k}$ along x, y, z axes.

Position vector: $\mathbf{r} = x \hat{i} + y \hat{j} + z \hat{k}$.

Malayalam support: Scalar-മാത്ര മാനിത (ഉദാ: mass). Vector-
മാത്ര മാനിത-ഉപദിഷ്ഠിത (ഉദാ: velocity). Unit vectors $\hat{i}, \hat{j}, \hat{k}$ axes-
അനുസരിച്ചുള്ള യൂണിറ്റ് വെക്ടറുകൾ.

Magnitude and Unit Vector

Magnitude: $|\mathbf{v}| = \sqrt{x^2 + y^2 + z^2}$ for $\mathbf{v} = x \hat{i} + y \hat{j} + z \hat{k}$.

Unit vector: $\hat{v} = \mathbf{v} / |\mathbf{v}|$ (direction of $\mathbf{v}$).

Example: $\mathbf{v} = 3\hat{i} + 4\hat{j}$. $|\mathbf{v}| = 5$. $\hat{v} = (3/5)\hat{i} + (4/5)\hat{j}$.

Vector Addition, Subtraction, Scalar Multiplication

- **Addition:** $\mathbf{v} + \mathbf{w} = (v_1 + w_1)\hat{i} + (v_2 + w_2)\hat{j} + (v_3 + w_3)\hat{k}$
- **Subtraction:** $\mathbf{v} - \mathbf{w} = (v_1 - w_1)\hat{i} + (v_2 - w_2)\hat{j} + (v_3 - w_3)\hat{k}$
- **Scalar multiplication:** $c \cdot \mathbf{v} = (c v_1)\hat{i} + (c v_2)\hat{j} + (c v_3)\hat{k}$

Example: $\mathbf{v} = 2\hat{i} + 3\hat{j}$, $\mathbf{w} = 1\hat{i} - 2\hat{j}$. $\mathbf{v} + \mathbf{w} = 3\hat{i} + 1\hat{j}$, $\mathbf{v} - \mathbf{w} = 1\hat{i} + 5\hat{j}$, $3\mathbf{v} = 6\hat{i} + 9\hat{j}$.

Dot Product

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}| |\mathbf{w}| \cos \theta \quad \cdot \quad \mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3$$

- **Angle:** $\cos \theta = (\mathbf{v} \cdot \mathbf{w}) / (|\mathbf{v}| |\mathbf{w}|)$
- **Perpendicular:** $\mathbf{v} \perp \mathbf{w} \Leftrightarrow \mathbf{v} \cdot \mathbf{w} = 0$

Example: $\mathbf{v} = 2\hat{i} + 3\hat{j}$, $\mathbf{w} = 1\hat{i} - 2\hat{j}$. $\mathbf{v} \cdot \mathbf{w} = 2(1) + 3(-2) = 2 - 6 = -4$.
 $|\mathbf{v}| = \sqrt{13}$, $|\mathbf{w}| = \sqrt{5}$, $\cos \theta = -4 / (\sqrt{13} \cdot \sqrt{5}) \approx -0.496$.

Cross Product

$$\mathbf{v} \times \mathbf{w} = |\mathbf{v}| |\mathbf{w}| \sin \theta \cdot \hat{n} \quad \cdot \quad \mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2) \hat{i} - (v_1 w_3 - v_3 w_1) \hat{j} + (v_1 w_2 - v_2 w_1) \hat{k}$$

- **Parallel:** $\mathbf{v} \parallel \mathbf{w} \Leftrightarrow \mathbf{v} \times \mathbf{w} = \mathbf{0}$

Example: $\mathbf{v} = 2\hat{i} + 3\hat{j}$, $\mathbf{w} = 1\hat{i} - 2\hat{j}$. $\mathbf{v} \times \mathbf{w} = (3 \cdot 0 - 0 \cdot (-2)) \hat{i} - (2 \cdot 0 - 0 \cdot 1) \hat{j} + (2 \cdot (-2) - 3 \cdot 1) \hat{k} = -7\hat{k}$.

 **Engineering Application:** Cross product gives torque ($\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$) and is used in magnetism and rotational dynamics.



Vector Calculator

Enter components of two vectors (a, b) and compute dot product, cross product, magnitudes, and angle.

a_1

2

a_2

3

a_3

0

b_1

1

b_2

-2

b_3

0

$a \cdot b = -4.000$ | $a \times b = (0.000, 0.000, -7.000)$ | $|a| = 3.606$ | $|b| = 2.236$ | $\theta = 119.74^\circ$



Formula Bank

$$z = a + ib, i^2 = -1$$

Complex number

$$|z| = \sqrt{a^2 + b^2}$$

Modulus

$$\theta = \tan^{-1}(b/a)$$

Argument

$$z = r(\cos \theta + i \sin \theta)$$

Polar form

$$P(E) = n(E)/n(S)$$

Probability

$$\bar{x} = \Sigma x / n$$

Mean

$$\int x^n dx = x^{n+1}/(n+1) + C$$

Power rule

$$\int u dv = uv - \int v du$$

Integration by parts

$$\int_a^b f(x) dx = F(b) - F(a)$$

Definite integral

$$I.F. = e^{\int P dx}$$

Linear D.E.

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3$$

Dot product

$$\mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2) \hat{i} - \dots$$

Cross product



Diagram Library for Rapid Revision



Argand Plane

Module 1 — Argand Plane showing complex number representation.



Probability Tree

Visualise outcomes for coin/die experiments via tree diagrams (conceptual).



Vector Geometry

Visualise dot and cross products in 3D space. Module 4



Self-Learning Activities (Official)

Week 1-4 · 16 h

Module I — Complex Numbers

- Find conjugate, modulus, amplitude, polar form
- Problems on equality, addition, subtraction
- Multiply and divide complex numbers
- Applications in engineering areas

Week 5-8 · 15 h

Module II — Probability & Statistics

- Problems with coins, die, cards
- Find mean, median, mode
- Frequency table problems
- Applications in engineering

Week 9-12 · 14 h

Module III — Integral Calculus

- Find integrals of functions
- Evaluate definite integrals
- Solve first-order differential equations
- Applications in engineering

Week 13-15 · 15 h

Module IV — Numerical Methods & Vectors

- Interpolation and Simpson's 1/3rd rule problems
- Vector addition, subtraction, scalar multiplication
- Dot product and cross product
- Area and work done problems

Total self-learning hours: 60 (as per syllabus)



Assessment Methodology

CIE / ESE Breakup

Mode	Attendance	CA1	CA2	CA3	CA4	Series Exam (2 modules)	Series Exam (2 modules)	ESE
Duration	—	2h	2h	2h	2h	2h	2h	2.5h
Exam mark	—	15	15	15	15	15	15	60
Converted to	5	15				20		60
Total	40 (CIE) + 60 (ESE)							

Series Examination Pattern

Duration: 2 hours

Total marks: 50 (converted to 20)

- Part A: 2×1 mark
- Part B: 2×3 marks
- Part C: 2×7 marks (with choice)

End Semester Examination

Duration: 2.5 hours

Total marks: 60

- Part A: 1 mark $\times$ 12 questions (2 per module)
- Part B: 3 marks $\times$ 6 questions (2 out of 3 per module)
- Part C: 7 marks $\times$ 6 questions (1 set with choice per module)

? Module-wise Questions with Answers

Module I — Complex Numbers

Q1: Find the modulus and argument of $z = 1 + i$.▼

$$|z| = \sqrt{(1^2+1^2)} = \sqrt{2}. \theta = \tan^{-1}(1/1) = 45^\circ.$$

Q2: Express $z = 2 - 2i$ in polar form.▼

$$r = \sqrt{(4+4)} = 2\sqrt{2}. \theta = -45^\circ. z = 2\sqrt{2}(\cos(-45^\circ) + i \sin(-45^\circ)).$$

Q3: If $z_1 = 1+2i$ and $z_2 = 3-i$, find $z_1 \times z_2$.▼

$$(1+2i)(3-i) = 3-i+6i-2i^2 = 3+5i+2 = 5+5i.$$

Q4: What is the conjugate of $3 - 4i$?▼

Conjugate is $3 + 4i$.

Module II — Probability & Statistics

Q1: A coin is tossed twice. Find the probability of getting exactly one head.▼

$$S = \{HH, HT, TH, TT\}. n(S)=4. E = \{HT, TH\}. n(E)=2. P = 2/4 = 1/2.$$

Q2: Find the mean of 3, 5, 7, 7, 9.▼

$$\text{Mean} = (3+5+7+7+9)/5 = 31/5 = 6.2.$$

Q3: What is the median of 2, 4, 6, 8?▼

$$n=4 \text{ (even)}. \text{Median} = (4+6)/2 = 5.$$

Q4: A die is rolled. Find P(prime number).▼

$$S = \{1,2,3,4,5,6\}. \text{Primes} = \{2,3,5\}. P = 3/6 = 1/2.$$

Module III — Integral Calculus

Q1: Evaluate $\int 3x^2 \, dx$.▼

$$\int 3x^2 \, dx = 3 \cdot (x^3/3) + C = x^3 + C.$$

Q2: Evaluate $\int_0^1 x \, dx$.▼

$$[x^2/2]_0^1 = 1/2 - 0 = 1/2.$$

Q3: Solve $dy/dx = 2x$.▼

$$dy = 2x \, dx \Rightarrow \int dy = \int 2x \, dx \Rightarrow y = x^2 + C.$$

Q4: Find the order and degree of $d^2y/dx^2 + 3(dy/dx) + 2y = 0$.▼

Order = 2 (highest derivative), Degree = 1 (exponent of highest derivative).

Module IV — Numerical Methods & Vector Algebra

Q1: Given (1,2), (2,3), (3,5), find y at x=2.5 using Lagrange's method.▼

$$L(2.5) = 2 \cdot \ell_0 + 3 \cdot \ell_1 + 5 \cdot \ell_2 = 3.875 \text{ (as worked in section).}$$

Q2: Approximate $\int_0^1 x^2 \, dx$ using Simpson's 1/3rd with n=2.▼

$$h = 0.5. \, f_0=0, \, f_1=0.25, \, f_2=1. \, \approx (0.5/3)[0+1+4(0.25)] = (1/6)[1+1] = 1/3 = 0.3333.$$

Q3: Find the magnitude of $\mathbf{v} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$.▼

$$|\mathbf{v}| = \sqrt{9+16} = 5.$$

Q4: Compute the dot product of $\mathbf{a} = (2,3)$ and $\mathbf{b} = (1,-2)$.▼

$$\mathbf{a} \cdot \mathbf{b} = 2(1)+3(-2) = 2-6 = -4.$$



Practice Model Paper

Note: This is a practice paper based on the official pattern, not an official SITTTTR model question paper.

Course 2001A — Engineering Mathematics – I

Duration: 2.5 hours · **Total marks:** 60

All modules are covered. Answer all parts.

Part A (1 mark each — 12 questions, 2 per module)

- **Q1.** What is the conjugate of $2 + 5i$? (M1)

- **Q2.** Find $|3 - 4i|$. (M1)
- **Q3.** A coin is tossed once. What is $P(H)$? (M2)
- **Q4.** Define sample space. (M2)
- **Q5.** $\int 2x \, dx = ?$ (M3)
- **Q6.** What is the order of $d^2y/dx^2 + y = 0$? (M3)
- **Q7.** What is a unit vector? (M4)
- **Q8.** When are two vectors perpendicular? (M4)
- **Q9-12.** (Additional questions covering all modules)

Part B (3 marks each — 6 questions, 2 out of 3 per module)

- **M1:** Express $1 - i$ in polar form. *OR* Find modulus and argument of $-1 + i$.
- **M2:** Two coins are tossed. Find $P(\text{at least one head})$. *OR* Find mean/median/mode of 2,3,3,5,6.
- **M3:** Evaluate $\int x \sin x \, dx$. *OR* Solve $dy/dx = 2xy$.
- **M4:** Given (1,2), (2,4), (3,6), find y at $x=2.5$ using Lagrange. *OR* Find $a \cdot b$ and $a \times b$ for $a=(1,2,0)$, $b=(0,1,1)$.

Part C (7 marks each — 6 questions, 1 set with choice per module)

- **M1:** If $z_1 = 2+3i$ and $z_2 = 1-2i$, find z_1+z_2 , z_1-z_2 , $z_1 \cdot z_2$, and z_1/z_2 . *OR* Express $z = 2+2i$ in polar form and find z^3 .
- **M2:** A card is drawn from a deck. Find $P(\text{heart})$, $P(\text{ace})$, $P(\text{heart or ace})$. *OR* Given frequency distribution, compute mean, median, mode.
- **M3:** Evaluate $\int_0^2 (x^2+2x) \, dx$. Solve $dy/dx + 2y = e^x$. *OR* Find area under $y = x^2$ from $x=0$ to $x=2$.
- **M4:** Use Simpson's 1/3rd to approximate $\int_0^2 x^3 \, dx$ with $n=4$. Find $a \times b$ for $a=(2,1,0)$, $b=(0,3,4)$. *OR* Lagrange interpolation problem with 4 points.

Quick Revision

Module I — Complex Numbers

- $z = a + ib$, $i^2 = -1$
- $|z| = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}(b/a)$
- Polar: $z = r(\cos \theta + i \sin \theta)$
- Operations: add real/imag separately

Module II — Probability & Stats

- $P(E) = n(E)/n(S)$
- Mean = $\Sigma x/n$, Median = middle, Mode = most frequent
- Sample space for coins/dice/cards

Module III — Integral Calculus

- $\int x^n dx = x^{n+1}/(n+1) + C$
- $\int u dv = uv - \int v du$
- Definite: $\int_a^b = F(b) - F(a)$
- D.E.: variable separable & linear

Module IV — Num Methods & Vectors

- Lagrange: $L(x) = \Sigma y_i \cdot \ell_i(x)$
- Simpson's: $h/3 [f_0 + f_n + 4\Sigma \text{odd} + 2\Sigma \text{even}]$
- $a \cdot b = |a||b| \cos \theta$
- $a \times b = |a||b| \sin \theta \cdot \hat{n}$

⚠ Common Mistakes — Exam Checklist

- ✓ In complex division, multiply by conjugate of denominator
- ✓ In probability, check if order matters (permutations vs combinations)
- ✓ In integration, always add +C for indefinite integrals
- ✓ In Simpson's rule, n must be even
- ✓ In vectors, cross product gives a vector, dot product gives a scalar
- ✓ In differential equations, identify order and degree correctly

Glossary

Term	Meaning
Complex number	$z = a + ib$, where $i^2 = -1$
Modulus	$ z = \sqrt{a^2+b^2}$, distance from origin
Argument	$\theta = \tan^{-1}(b/a)$, angle with real axis
Polar form	$z = r(\cos \theta + i \sin \theta)$
Sample space	Set of all possible outcomes
Probability	$P(E) = n(E)/n(S)$
Mean	Average = $\Sigma x/n$
Median	Middle value of ordered data
Mode	Most frequent value
Integration	Reverse process of differentiation
Definite integral	$\int_a^b f(x) dx = F(b) - F(a)$
Differential equation	Equation involving derivatives
Interpolation	Estimating unknown values from known data
Lagrange's method	Polynomial interpolation through given points
Simpson's 1/3rd rule	Numerical integration using parabolic arcs
Scalar	Quantity with only magnitude
Vector	Quantity with magnitude and direction
Dot product	Scalar product: $a \cdot b = a b \cos \theta$
Cross product	Vector product: $a \times b = a b \sin \theta \cdot \hat{n}$

Learning Resources

Textbooks		
Sl. No.	Title	Author / Publication
1	Mathematics for Industrial Engineering	SITTTR Kalamasser

References		
Sl. No.	Title	Author / Publication
1	Higher Engineering Mathematics	B. S. Grewal, Khanna Publishers
2	Higher Engineering Mathematics	B. V. Ramana, Tata McGraw-Hill
3	A text book of Engineering Mathematics	N. P. Bali & Manish Goyal, Laxmi Publications
4	Schau's outlines of Discrete Mathematics (3rd Ed.)	Seymour Lipschutz & Marc Lipson, McGraw Hill
5	Schau's outlines of Probability (3rd Ed.)	Seymour Lipschutz & Marc Lipson, McGraw Hill
6	Schau's outlines of Statistics (6th Ed.)	Dr. Murray R. Spiegel & Larry J. Stephens, McGraw Hill

Web resources: None officially listed in the syllabus.